



Safe Fuel Delivery at Engine Dyno

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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1.0 Introduction

The Dyno Engine Test room, a common shop at dealers, is where engines are tested to see whether they are working properly or not. To perform a test, the engine fuel line should be filled with fuel and then pressure should be increased. Normally, a manual process is in place where the fuel transfer pump is used with the help of a battery. 24 units is the voltage necessary to run the fuel transfer pump, but in many countries, including Turkey, 220 units is the standard voltage of electric sockets, which is why a battery must be used.

One of our technicians at Borusan Makina ve Güç Sistemleri has found a new way to perform this process by using a converter to decrease the voltage, eliminating the use of the battery. This new process is easier and safer to perform than the old one, because it does not require the technician to be in the room while the test is being performed.

2.0 Best Practice Description

In this new process the battery is replaced with a converter that decreases voltage from 220 units to 24 units. The converter is controlled with a key that you press to start the process. Electricity is sent to the converter. The converter converts electricity to the necessary voltage and sends the electricity to the gas pump. The gas pump then fills the engine fuel line with fuel.

3.0 Implementation Steps

1. Connect the fuel transfer pump and converter with the cable.



In these photos the connection cable can not be seen since it passes under the floor.

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2. Connect the converter and key. This completes the process and the engine is ready to be tested.
3. If you press the “FUEL” button, the fuel line will be filled with fuel from the gas pump.



4.0 Benefits

- Reduces engine preparation labor hours since one technician can test the engine;
- Creates a safer working environment since no technician needs to be in the test room;
- Reduces electricity cost since the battery is not needed.

5.0 Resources Required

Item	Cost
Converter	\$60
5 Meter Cable	\$5
Key	\$4
Total	\$69

6.0 Supporting Attachments / References

N/A.

7.0 Related Best Practices

N/A.

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